2026 BROCK AI DAY · EXPLAINABLE AI / DEEP LEARNING

Are Language Models Converging on Shared Structures?

Mechanistic Interpretability, the Linear Representation Hypothesis, and the Platonic Representation Hypothesis

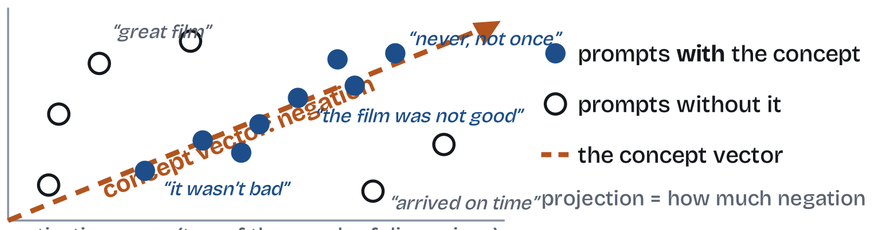
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1 Concepts are vectors

Linear Representation Hypothesis (Park, Choe & Veitch, 2024): a model stores each concept it learns, such as negation or sentiment, as a *vector* in its activation space. The more strongly a prompt expresses the concept, the further it lands along that vector, so measuring along it tells you how much of the concept is present. Each layer has its own version of the vector, and concepts can overlap when they share dimensions (superposition; Elhage et al., 2022).

Mechanistic interpretability reads a model's internal activations to find what it represents and how, rather than judging it by its outputs alone. It needs the model's weights, so it is only possible with open models.

WHERE THE VECTORS LIVE The **residual stream**: one running vector per token that every layer takes as input and modifies via attention and MLP. We read it at the *final token* of a passage, at any layer: that position has attended to every earlier token, so its vector carries the whole passage, not the last word.



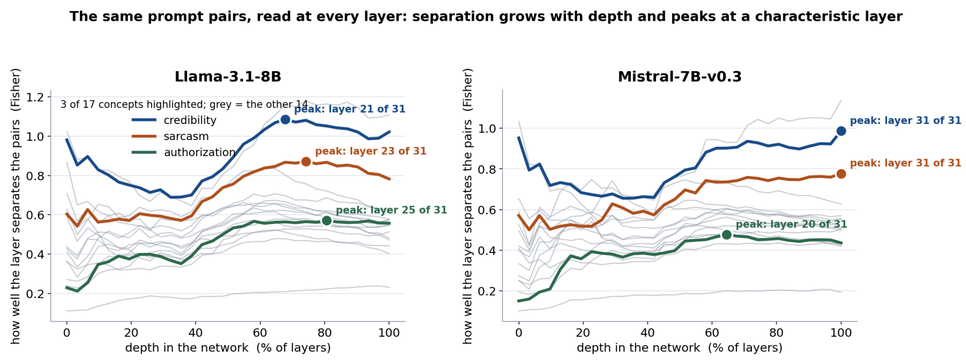
2 Extracting a concept vector from contrastive pairs

Pairs. For each concept, write passages on the same topic in two versions, one carrying the concept and one not. Here: 17 concepts, about 250 pairs each, written by 14 different LLMs. Run both versions through the model and record the residual stream at the final token of each passage, at every layer.

WITH: CREDIBILITY MARKERS
"Traditional vaccine development is a meticulous process, typically requiring 10–15 years..."

WITHOUT: SAME CLAIM, NO GROUNDING
"It's just common sense that a vaccine should take years to develop..."

Difference of means. At each layer, the mean activation of the *with* passages minus the mean of the *without* passages is that layer's concept vector (Zou et al., 2023; Rimsky et al., 2024). Score each layer by how well its vector separates the two sets. The peak layer is where the concept is most cleanly represented.



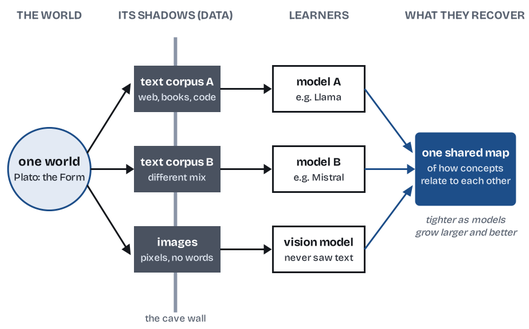
Separation at every layer for all 17 concepts in two models; credibility, sarcasm and authorization highlighted. Dots mark each concept's peak layer. Grammatical concepts such as negation and plurality peak early; sarcasm and authorization peak late.

3 Convergence: the Platonic Representation Hypothesis

Plato's cave: prisoners see only shadows cast by things they never see directly. The shadows change with the light; the things do not.

The hypothesis (PRH: Huh, Cheung, Wang & Isola, 2024): text, images and other data are shadows of one world, and models trained on different shadows, with different architectures and objectives, converge toward a shared model of that world, more so as they grow.

What converges is relations, not coordinates: which things are similar to which.
Prediction here: models that never shared a weight should agree on how concepts relate to each other.



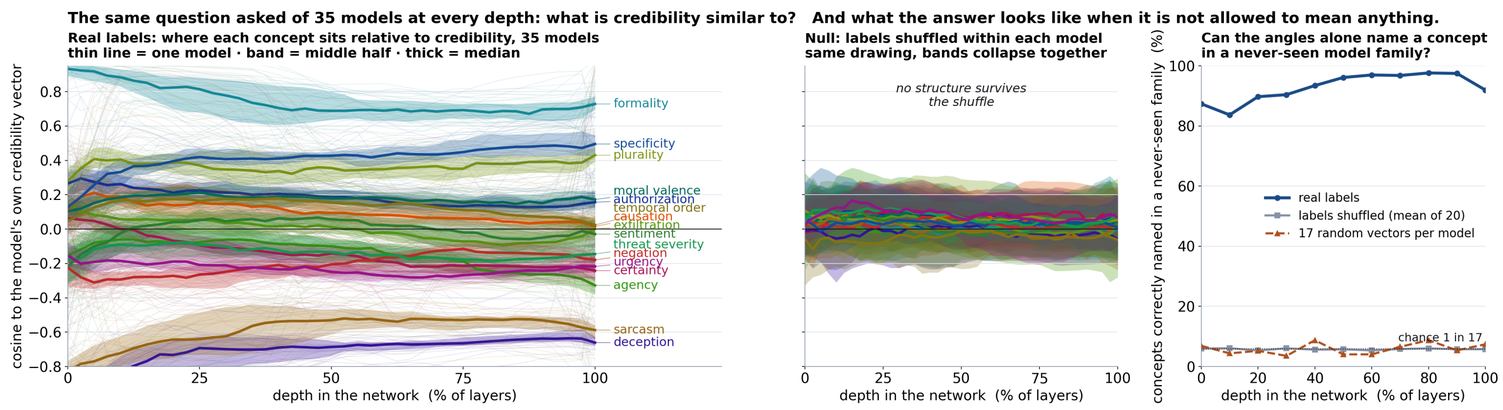
Different shadows, different learners, one structure recovered; the coincidence tightens with scale.

4 Our test: ask 35 models the same question

Each model's credibility vector lives in its own coordinate system, so two models' vectors cannot be laid on each other. What *can* be compared are the angles inside each model (relative representations: Moschella et al., 2023).

So ask 35 models, at every layer: **what is credibility similar to?** Formality sits closest in all of them, specificity next; sarcasm and deception sit far away. Wider bands mean looser agreement across models; all bands tighten from mid-stack on. The middle panel shows the shuffle.

The test that can fail: hold out a whole model family, describe each of its concept vectors only by its angles to the others. Mid-depth: 97% identified correctly. Shuffled labels or random vectors: 6%, which is chance.



What it might mean: the consolidation of meaning

Thirty-five models, built by different teams on different text, agree on how their concepts sit relative to one another. Each model's coordinates are its own; the angles between its concepts are not. That shared map is what the Platonic Representation Hypothesis predicts: different shadows, one structure recovered. And because the map is measurable, the hypothesis can be tested concept by concept and model by model. Making such claims testable is part of what mechanistic interpretability is for.

Is this the structure of language itself, or, as Plato would have it, the structure of the world it describes?

github.com/jamesrahenry/ai-day-2026

Poster source, figure code, full reading list and sources.

THIS WORK

Henry (2026a). Concept Allocation Zones. arXiv:2605.24856.

Henry (2026b). Geometric Evolution Maps. arXiv:2605.25848.

Henry (2026c). CAZ Validation. Zenodo 10.5281/zenodo.22016917.

Data: huggingface.co/datasets/james-ra-henry/Rosetta-Activations